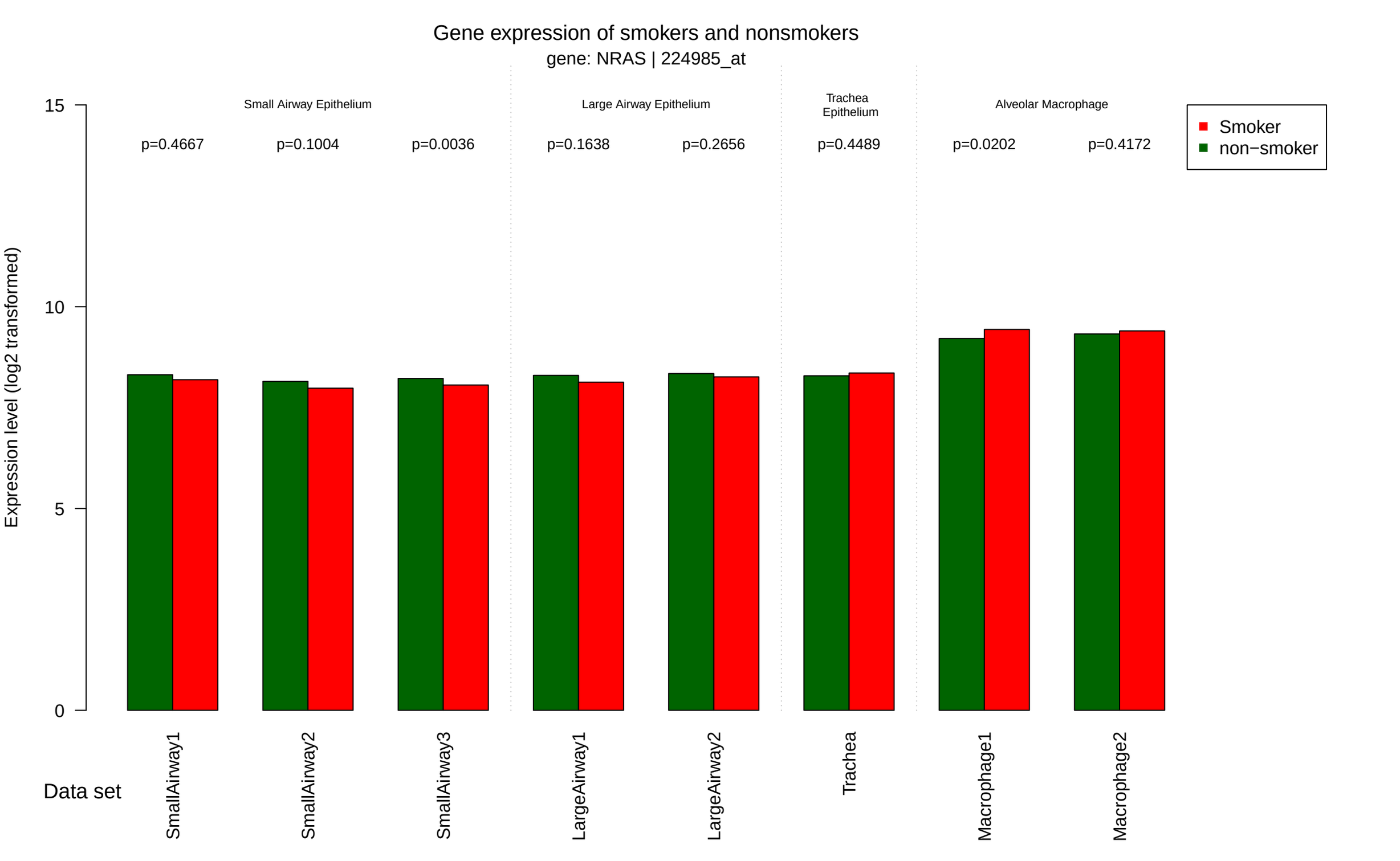
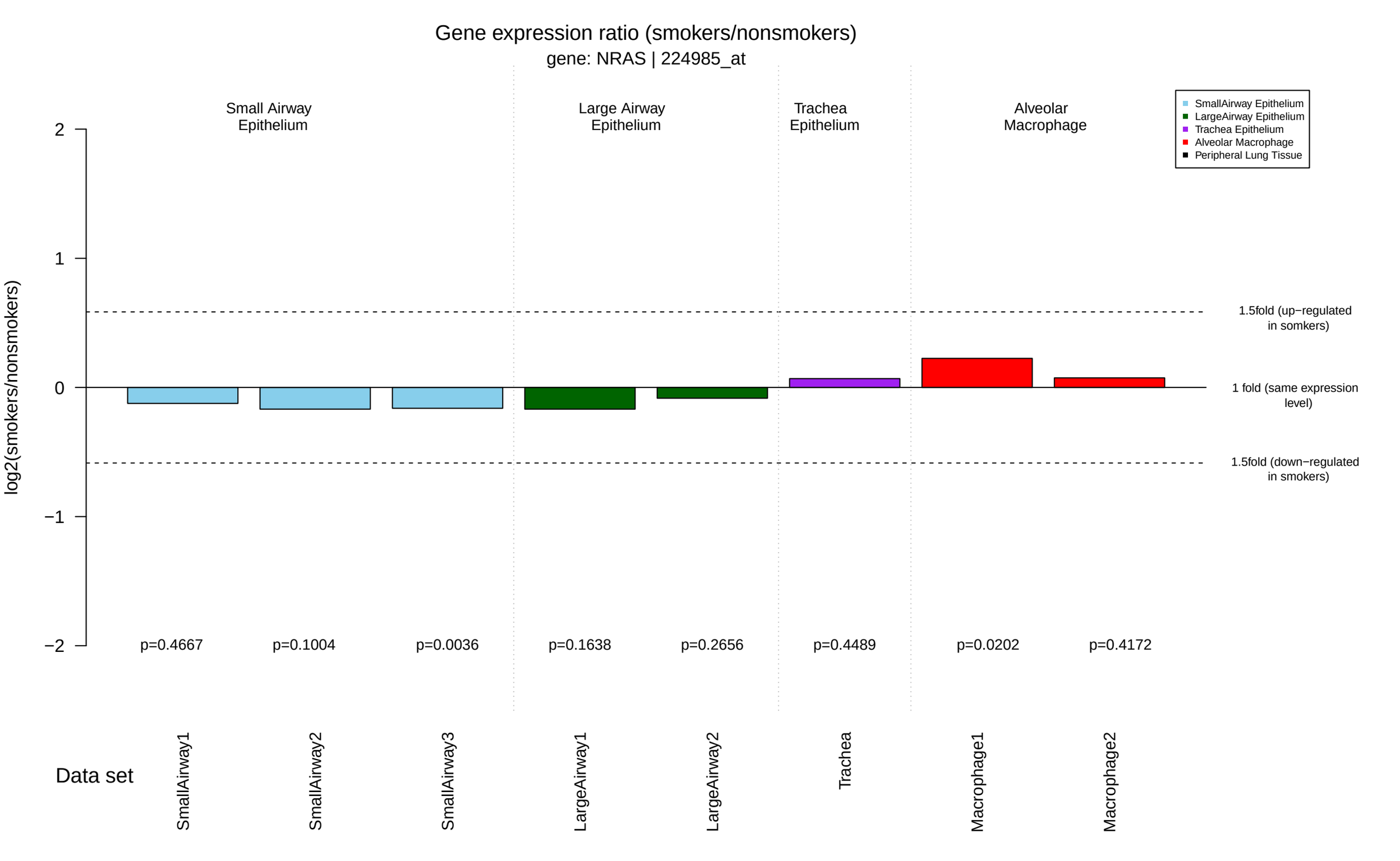
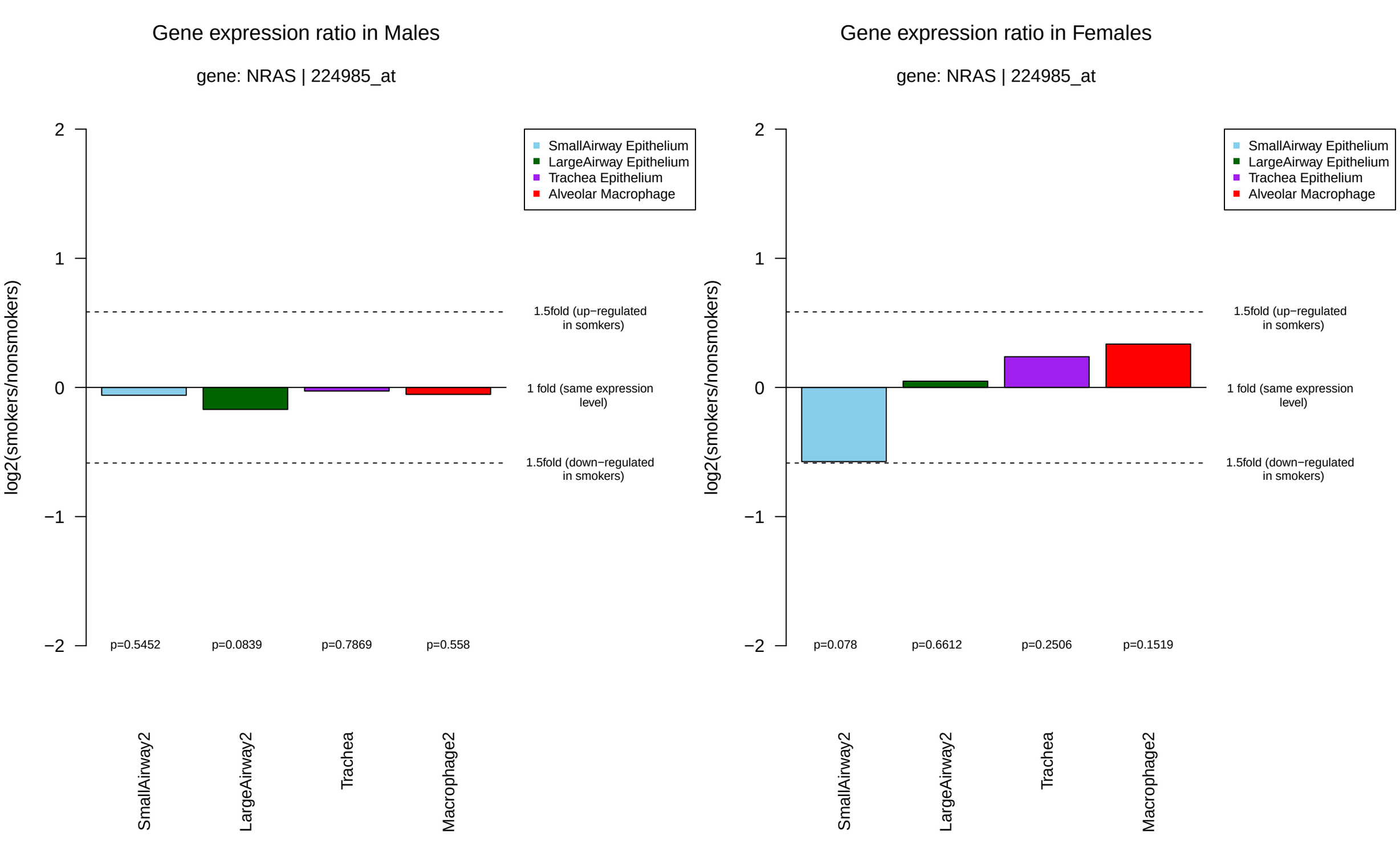
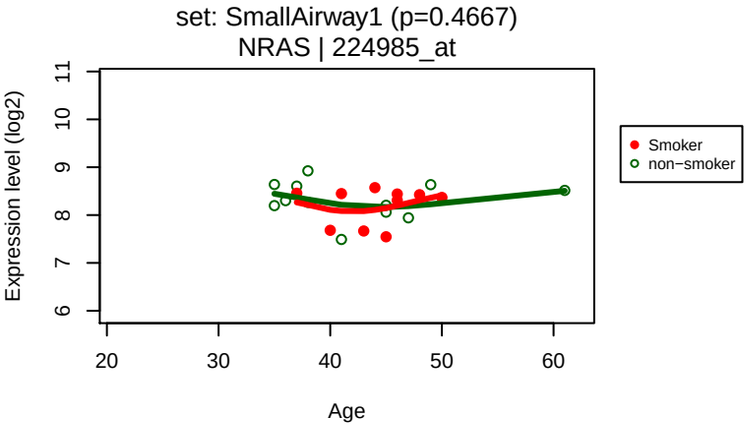
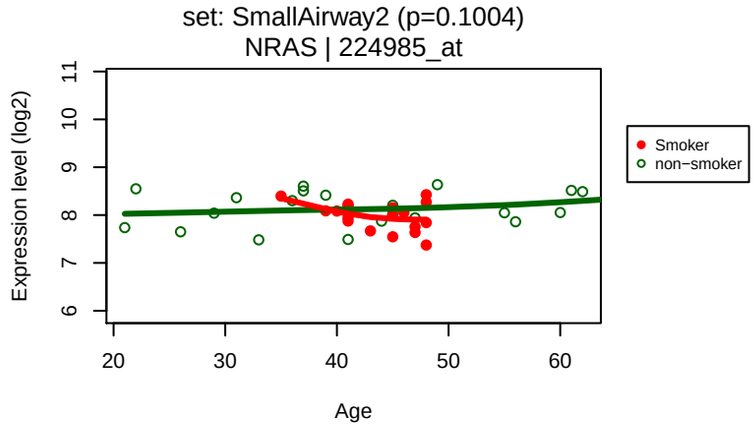
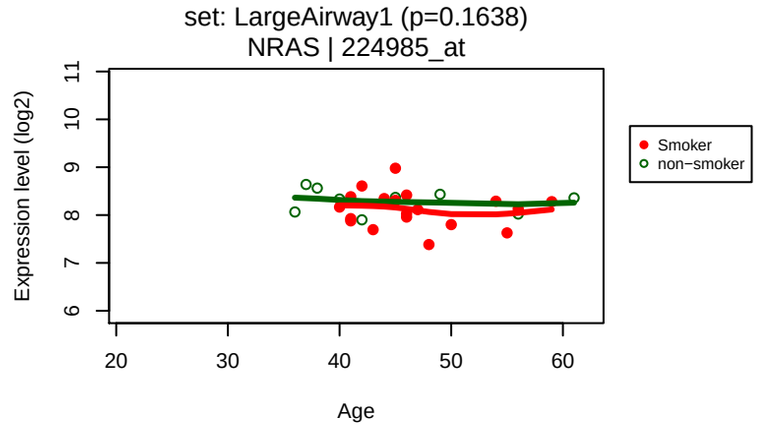

Gene expression of smokers and nonsmokers
gene: NRAS | 224985_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4667
p=0.1004
p=0.0036
p=0.1638
p=0.2656
p=0.4489
p=0.0202
p=0.4172
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

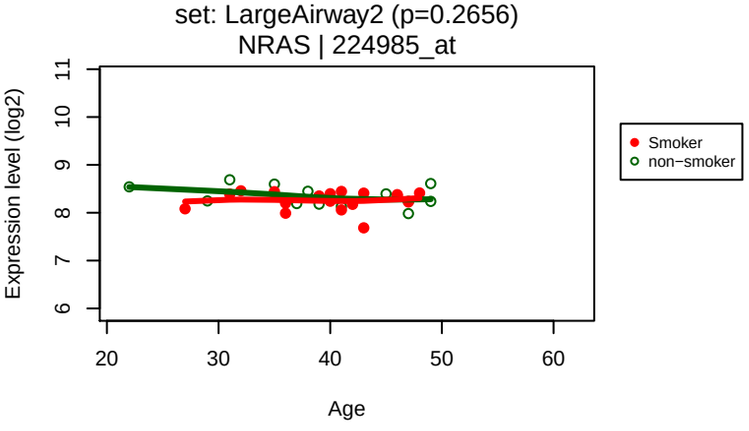

Gene expression ratio (smokers/nonsmokers)
gene: NRAS | 224985_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4667
p=0.1004
p=0.0036
p=0.1638
p=0.2656
p=0.4489
p=0.0202
p=0.4172
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

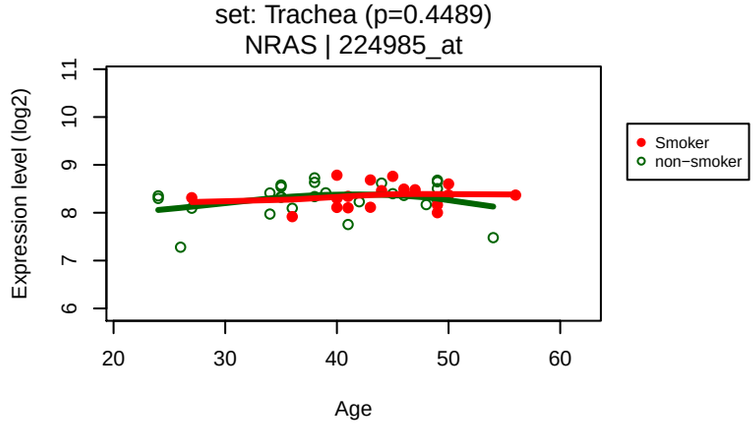

Gene expression ratio in Males
gene: NRAS | 224985_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5452
p=0.0839
p=0.7869
p=0.558
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NRAS | 224985_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.078
p=0.6612
p=0.2506
p=0.1519
SmallAirway2
LargeAirway2
Trachea
Macrophage2

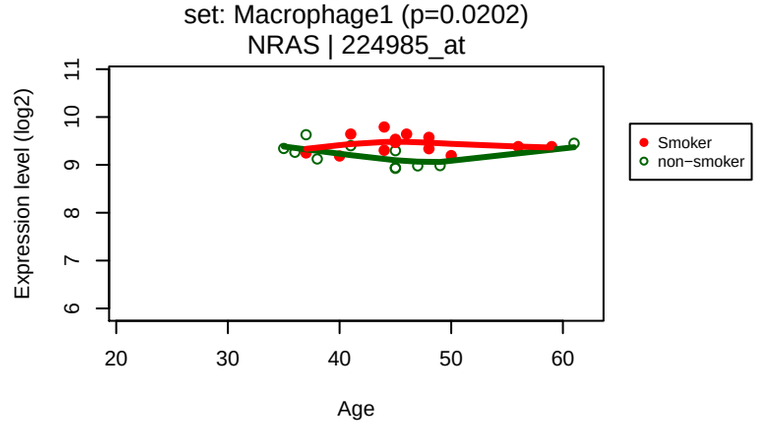

set: SmallAirway1 (p=0.4667)
NRAS | 224985_at
Expression level (log2)
Age
Smoker
non−smoker

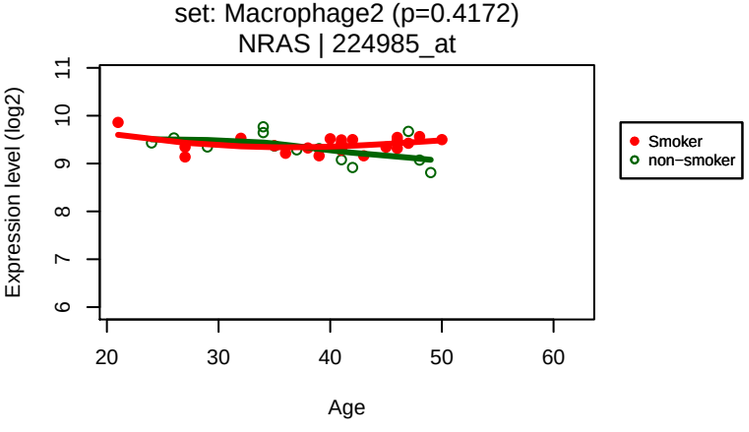

set: SmallAirway2 (p=0.1004)
NRAS | 224985_at
Expression level (log2)
Age
Smoker
non−smoker

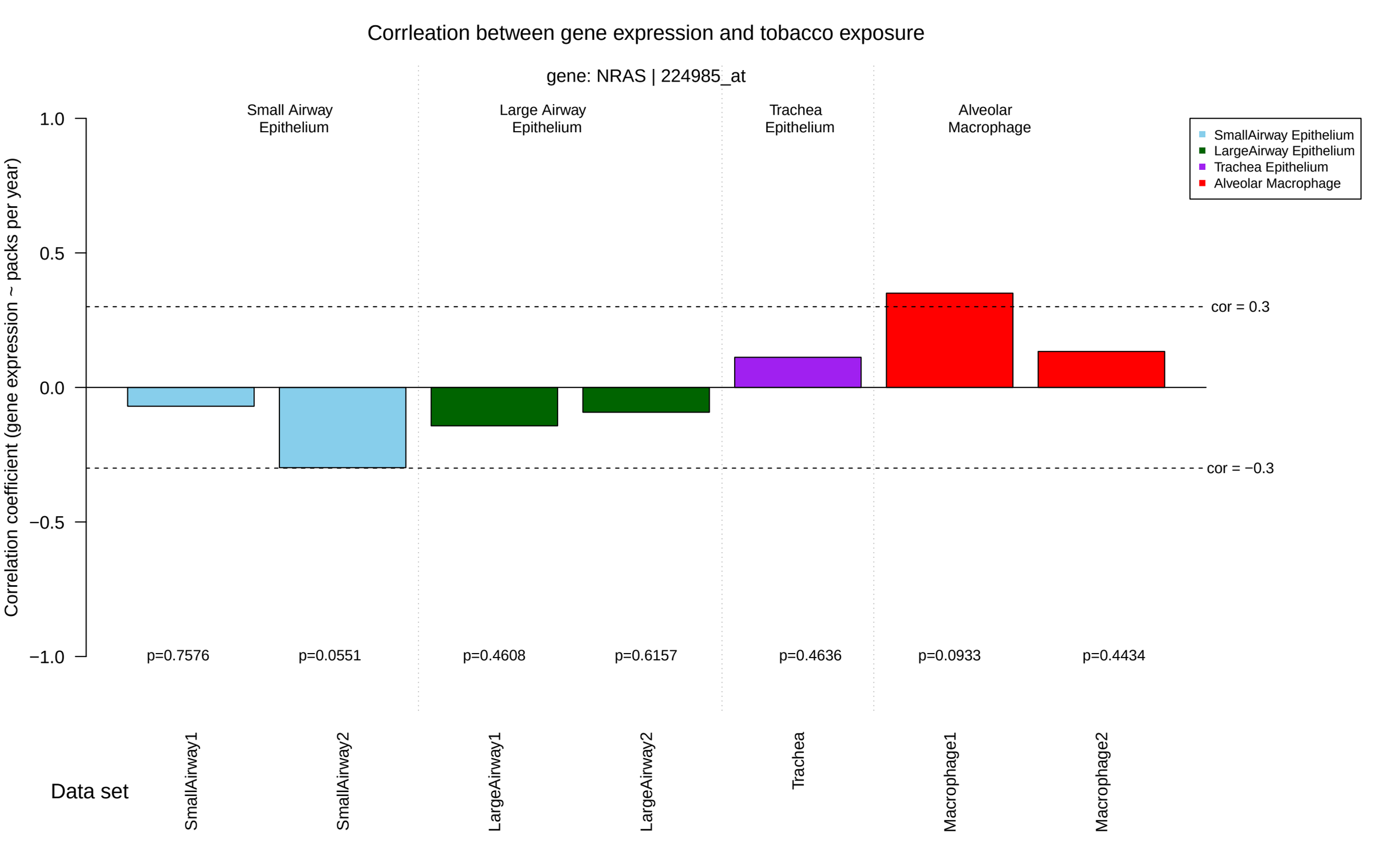

set: LargeAirway1 (p=0.1638)
NRAS | 224985_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2656)
NRAS | 224985_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4489)
NRAS | 224985_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0202)
NRAS | 224985_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4172)
NRAS | 224985_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NRAS | 224985_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7576
p=0.0551
p=0.4608
p=0.6157
p=0.4636
p=0.0933
p=0.4434
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2